

Chapter 1: The Water Cycle

The water cycle, also called the hydrologic cycle, describes how water moves continuously through the Earth and atmosphere. Driven by energy from the Sun, liquid water at the surface of oceans, lakes, and rivers evaporates into water vapor. Plants also release vapor through a process called transpiration. As the vapor rises, it cools and condenses into tiny droplets that form clouds. When the droplets grow heavy enough, they fall back to the surface as precipitation: rain, snow, sleet, or hail. Some of this water soaks into the ground to become groundwater, while the rest flows over land as runoff, returning to rivers and the ocean. The cycle has no beginning and no end; it simply repeats, redistributing fresh water around the planet and shaping weather and climate everywhere.

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Chapter 2: Photosynthesis

Photosynthesis is the process by which green plants, algae, and some bacteria convert light energy into chemical energy stored in sugars. Inside the leaves, structures called chloroplasts contain a green pigment named chlorophyll, which absorbs sunlight. Using that energy, the plant combines carbon dioxide drawn from the air through tiny pores called stomata with water pulled up from the roots. The reaction produces glucose, a sugar the plant uses for growth and energy, and releases oxygen as a by-product. This oxygen is essential for the breathing of most living things. Photosynthesis is the foundation of nearly every food chain on Earth, because it captures the Sun's energy in a form that other organisms can eat and use.

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Chapter 3: Newton's Laws of Motion

Sir Isaac Newton described three laws that explain how objects move. The first law, the law of inertia, states that an object stays at rest or keeps moving in a straight line at constant speed unless a net force acts on it. The second law explains how force changes motion: the acceleration of an object depends on the net force applied and its mass, summarized by the equation $\text{force} = \text{mass} \times \text{acceleration}$. The third law states that for every action there is an equal and opposite reaction, so when one object pushes on another, the second pushes back with the same strength in the opposite direction. Together these laws form the basis of classical mechanics and explain everything from a rolling ball to the orbit of a rocket.

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